



# Bike Sales Data Analysis

Comprehensive analysis to drive strategic sales growth and optimize business performance across key markets

**BY: Vishu Raj Srivastav**

**Harsh Upadhyay**

**Abhay Gupta**

**Vigyarth Bhadouriya**

## Analysis Overview

01

### Introduction & Purpose

Define objectives and scope of analysis

02

### Data Cleaning

Standardize and validate data quality

03

### Summary Statistics

Generate key metrics and visualizations

04

### Outlier Analysis

Identify and handle anomalous data points

05

### Feature Engineering

Create derived insights for analysis

06

### Key Insights

Extract actionable business intelligence

07

### Strategic Conclusions

Present executive-level recommendations

# Analysis Purpose

## Drive Sales Growth

Identify high-performing products and markets to maximize revenue opportunities

## Control Underperformance

Pinpoint weak areas requiring immediate attention and strategic intervention

## Optimize Operations

Streamline inventory management and resource allocation for peak efficiency

## DATA CLEANING PROCESS



Vector: VencorDataols.com

# Data Cleaning Process

## Remove Duplicates



Ensured all entries are unique to prevent skewed analysis and maintain data integrity

## Handle Missing Values



Applied median or mean imputation to fill gaps in critical columns like unit cost and price

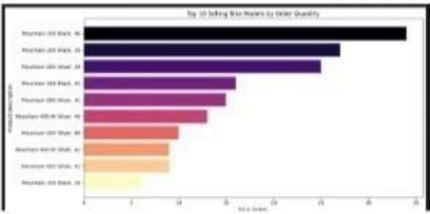
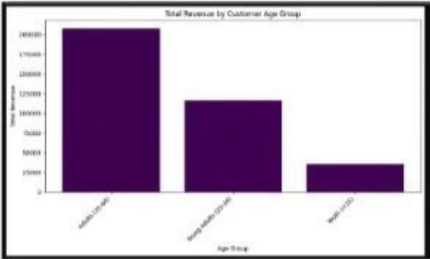
## Standardize Data



Normalized formats and scales across all variables for consistent analytical processing

✓ Result: Clean, standardized dataset ready for comprehensive statistical analysis

# Summary Statistics & Key Visualizations



## Revenue by Product

Categorical distribution showing top-performing motorcycle segments



## Sales by Demographics

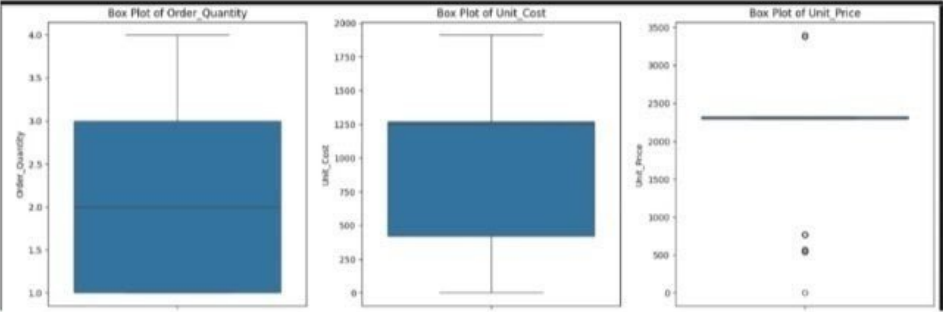
Age group analysis revealing primary customer segments and buying patterns



## Gender Distribution

Sales performance across male and female customer segments

# Outlier Detection



An outlier is a data point that differs significantly from other observations in your dataset. It represents unusually high or low values compared to the majority of the data.

### Detection Methods

Utilized boxplot analysis and scatter plot visualization to identify statistical anomalies in sales data

### Impact Assessment

Evaluated whether outliers represent exceptional sales opportunities or data quality issues requiring correction

# Feature Engineering Insights

1

## Profit Margin Analysis

Calculated profit-to-revenue ratios to identify most profitable motorcycle models and optimize pricing strategies

2

## Regional Product Mapping

Identified top-performing products per state to enable targeted regional marketing and inventory allocation

3

## Customer Segmentation

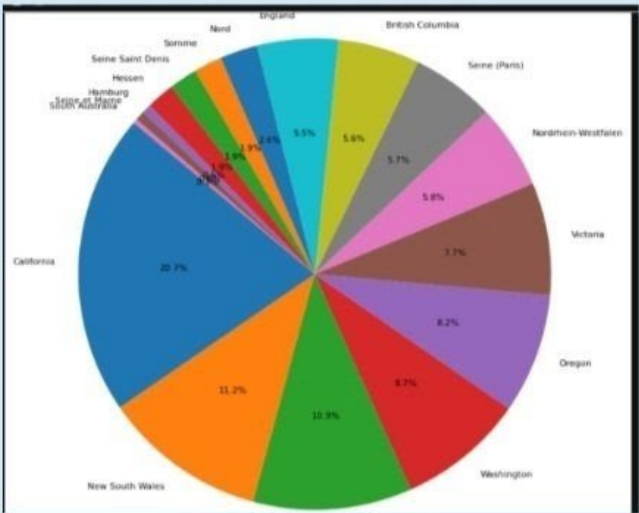
Categorized customers by age demographics to reveal highest-spending segments and tailor marketing approaches

These engineered features provide actionable insights for strategic decision-making and revenue optimization.

## Key Business Insights

### Revenue Hot Zones

Certain states consistently generate the highest revenue, representing critical markets for focused investment and expansion efforts



### High-Profit Models

Specific motorcycle models demonstrate exceptional profitability across multiple regions, warranting prioritized production and marketing allocation

### Prime Demographics

Adults aged 35-64 represent the largest buyer segment, with gender-specific preferences offering targeted campaign opportunities

### Seasonal Patterns

Distinct seasonal trends enable strategic promotion timing and optimized inventory planning for maximum sales impact



# Strategic Conclusions

## 1 Focus Marketing Investment

Concentrate promotional budgets and campaigns on high-performing motorcycle models in top revenue-generating states for maximum ROI

## 2 Optimize Customer Engagement

Track repeat customers and age-specific preferences to develop targeted upselling strategies and comprehensive loyalty programs

## 3 Strategic Inventory Management

Adjust stock levels based on state-specific demand patterns while reducing investment in consistently low-performing products

## 4 Seasonal Campaign Timing

Leverage identified seasonal trends to strategically time discount offers and new product launches for maximum market impact

## 5 Profit Margin Optimization

Phase out or significantly improve low-margin motorcycle models while doubling down on high-profitability segments